GT: routing strategies against mini and gpt-5.4

Generated 2026-10-04 14:21 EDT. Sources: /Users/elenabaron/MITxPro/RAGandContextEng/capstone/rag_and_context_engineering/wikipedia_retrieval_engine/Capstone Checkpoint 7.1/cost_efficiency/wikipedia_GT

Question set	Configurations	Runs	Question-runs	Baseline	Highest quality	Lowest list cost	Fastest
GT test set	6	10	190	7.1 final	7.1 final, gpt-5.4 (0.868)	6.1 survivor ($0.0485)	6.1 survivor (13.1 s)

How to read this report

Quality. pass = 1, partial = 0.5, fail = 0, averaged over question-runs (RAGAS judge).

List cost. every input token at the full price: the comparison of record. Billed cost charges cached input at the cached rate, so it falls when runs repeat the same questions.

Seconds. agent graph plus answer per question, judge excluded. Provider speed varies by hour; §6 shows how much.

Dominated. another configuration is at least as good and no more expensive (or slower), and better on one of the two.

Paired. each question's mean over its runs, configuration minus baseline; p from the Wilcoxon signed-rank test over the questions that changed.

Paths. Table: the TOC review flagged a table. Text, escalated: an assessment asked for more evidence. Text, ready: neither.

Loader warnings

- adoring_hopper: list cost repriced from its calls with the current price table ($4.9619 recorded -> $6.4348, +29.7%).

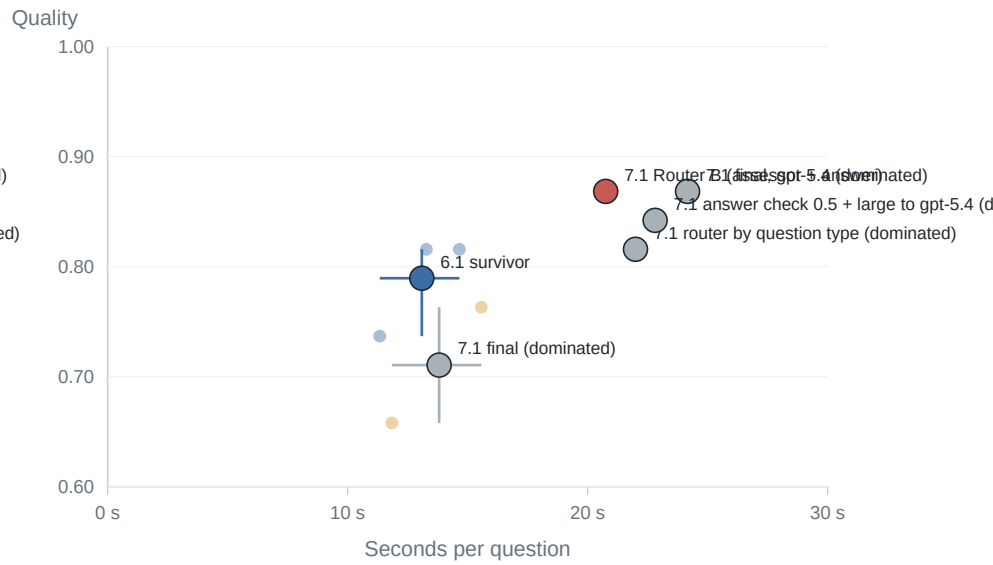
GT test set

6 configuration(s), 10 run(s), 190 question-runs. Baseline: 7.1 final (requested).

1. Frontier: quality against cost and time

Configuration	Runs (with telemetry)	Quality (run range)	Pass / partial / fail	List $/q (run range)	Billed $/q	Cached	Seconds mean / median	Calls/q	Input tokens/q	List $ per quality point	Dominated on cost by	Dominated on time by
6.1 survivor	3 (3)	0.789 (0.737-0.816)	40/10/7	$0.0485 ($0.0402-$0.0570)	$0.0350	35%	13.1 / 11.8	6.77	56.8K	$0.0615	-	-
7.1 final	3 (3)	0.711 (0.658-0.763)	37/7/13	$0.0502 ($0.0459-$0.0539)	$0.0348	38%	13.8 / 12.6	6.84	59.3K	$0.0706	6.1 survivor	6.1 survivor
7.1 final, gpt-5.4	1 (1)	0.868 (0.868)	16/1/2	$0.3387 ($0.3387)	$0.2787	20%	24.2 / 22.3	7.11	95.8K	$0.3900	7.1 Router B (assessor + answer)	7.1 Router B (assessor + answer)
7.1 router by question type	1 (1)	0.816 (0.816)	15/1/3	$0.3009 ($0.3009)	$0.2451	18%	22.0 / 19.4	7.74	94.9K	$0.3688	7.1 Router B (assessor + answer); 7.1 answer check 0.5 + large to gpt-5.4	7.1 Router B (assessor + answer)
7.1 Router B (assessor + answer)	1 (1)	0.868 (0.868)	15/3/1	$0.1417 ($0.1417)	$0.1013	42%	20.8 / 21.2	7.37	64.5K	$0.1631	-	-
7.1 answer check 0.5 + large to gpt-5.4	1 (1)	0.842 (0.842)	16/0/3	$0.1275 ($0.1275)	$0.0816	43%	22.8 / 15.7	8.95	71.2K	$0.1514	-	7.1 Router B (assessor + answer)





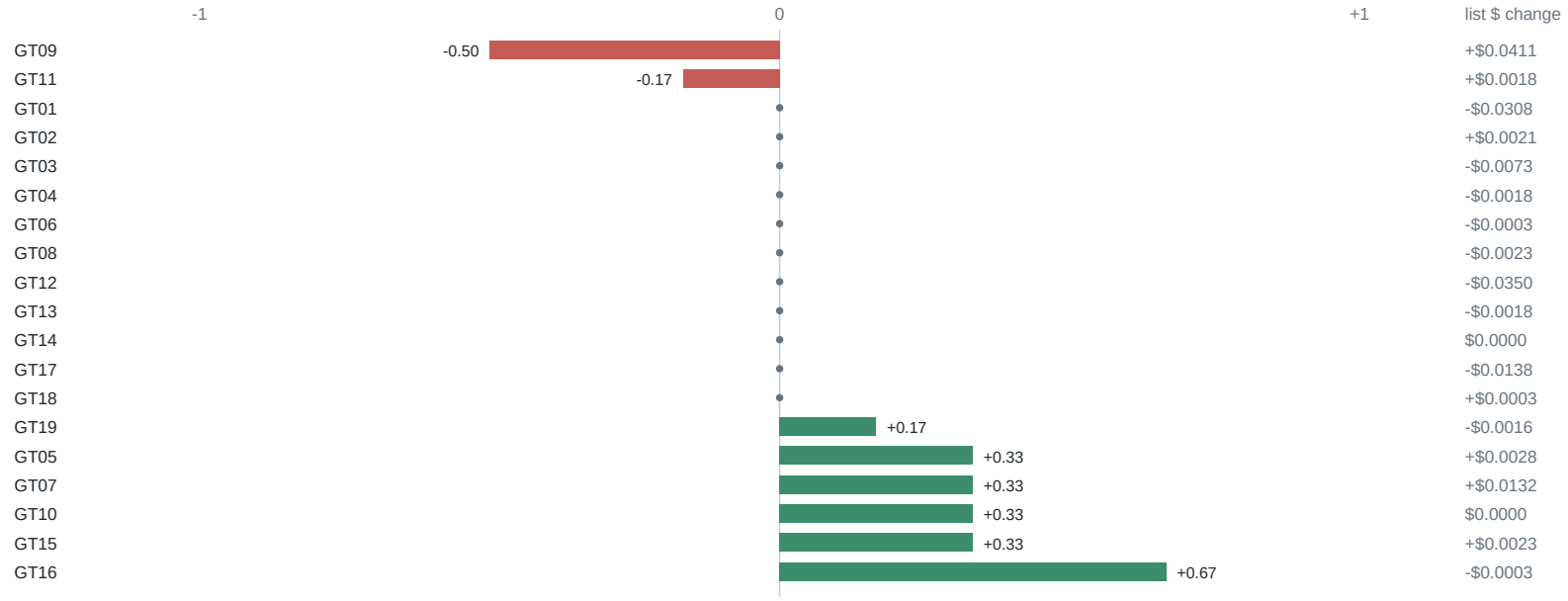
Small dots are runs; a large dot is the configuration mean, with lines across its run range. Gray: dominated.

2. Paired against the baseline (7.1 final)

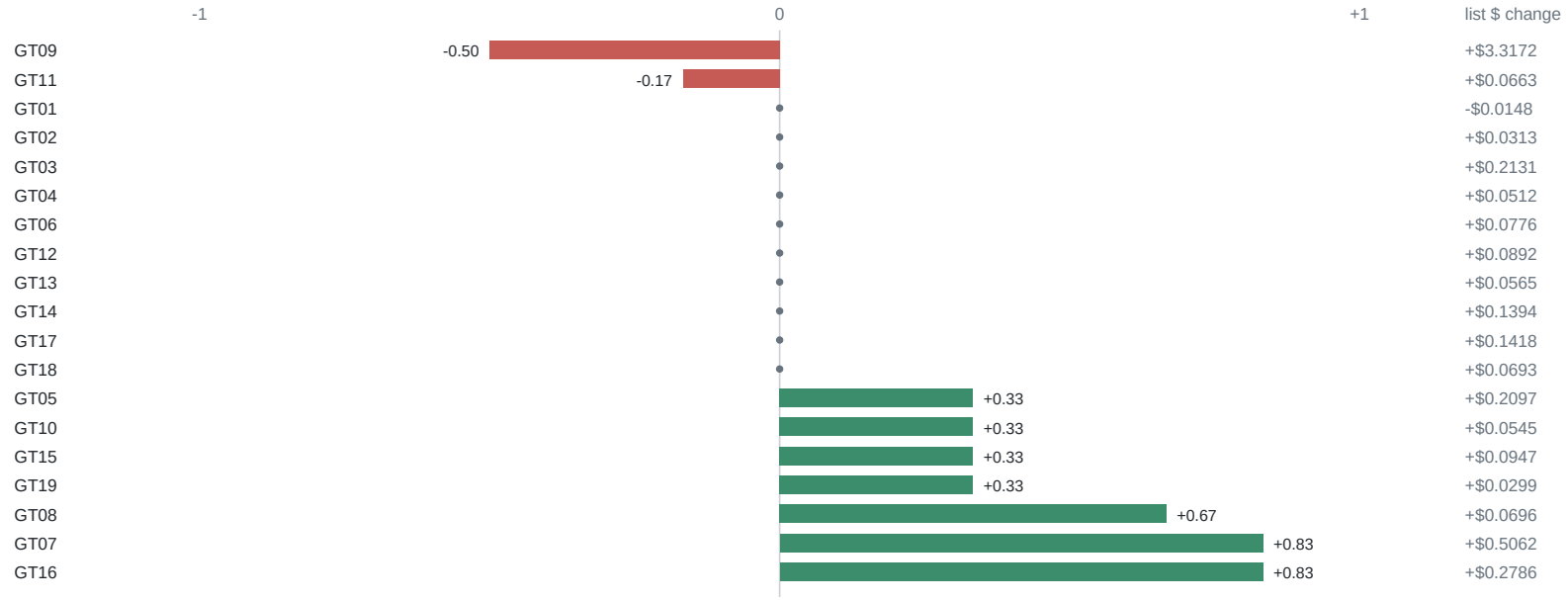
Configuration	Questions	Quality	Δ quality	Better / worse / same	p (quality)	Δ list $/q	p (cost)	Δ seconds	p (time)	Trade-off
6.1 survivor	19	0.711 -> 0.789	+0.079	6 / 2 / 11	0.188	-$0.0016 (-3%)	0.679	-0.7	0.061	dominates the baseline: at least as good, no more expensive
7.1 final, gpt-5.4	19	0.711 -> 0.868	+0.158	7 / 2 / 10	0.074	+$0.2885 (+575%)	<0.001	+10.3	<0.001	pays $0.1827 per question for each +0.10 quality
7.1 router by question type	19	0.711 -> 0.816	+0.105	6 / 3 / 10	0.133	+$0.2507 (+500%)	<0.001	+8.2	<0.001	pays $0.2382 per question for each +0.10 quality
7.1 Router B (assessor + answer)	19	0.711 -> 0.868	+0.158	7 / 2 / 10	0.074	+$0.0915 (+182%)	<0.001	+6.9	<0.001	pays $0.0579 per question for each +0.10 quality
7.1 answer check 0.5 + large to gpt-5.4	19	0.711 -> 0.842	+0.132	6 / 2 / 11	0.031	+$0.0773 (+154%)	0.196	+9.0	0.045	pays $0.0588 per question for each +0.10 quality

Per-question change in quality, worst first (right column: change in list cost):

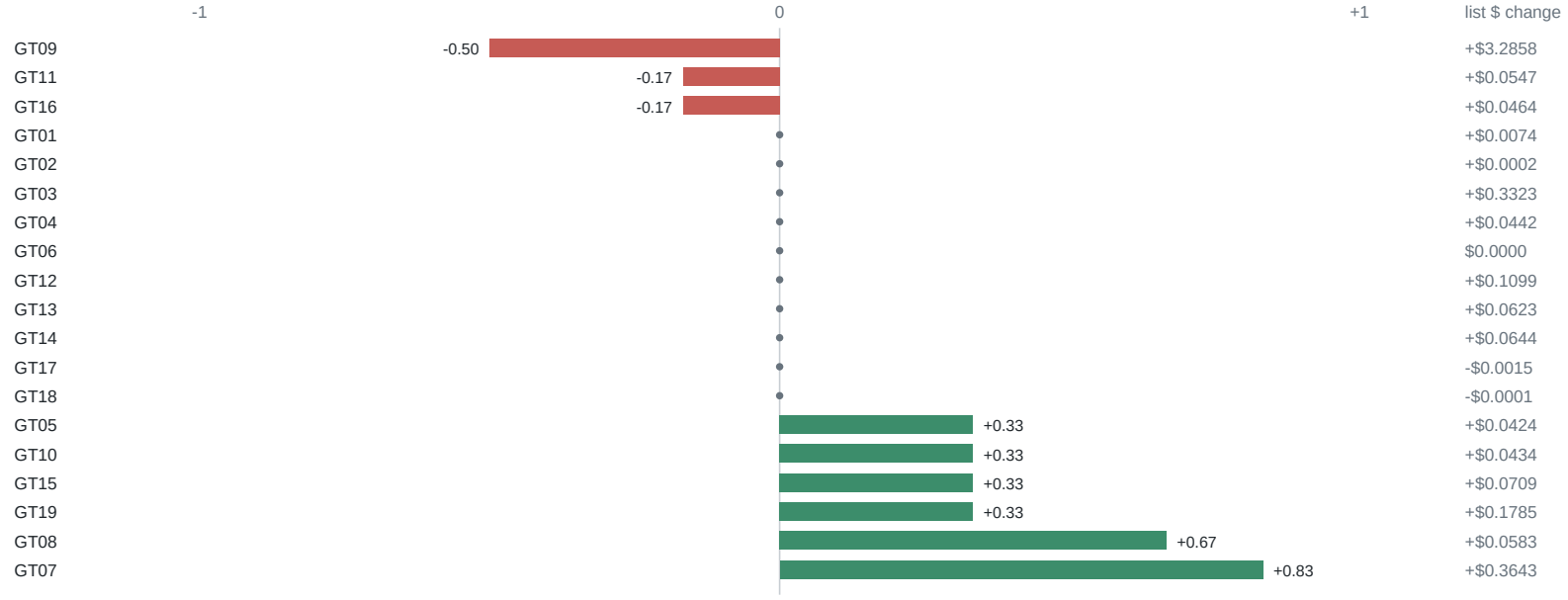
6.1 survivor against 7.1 final



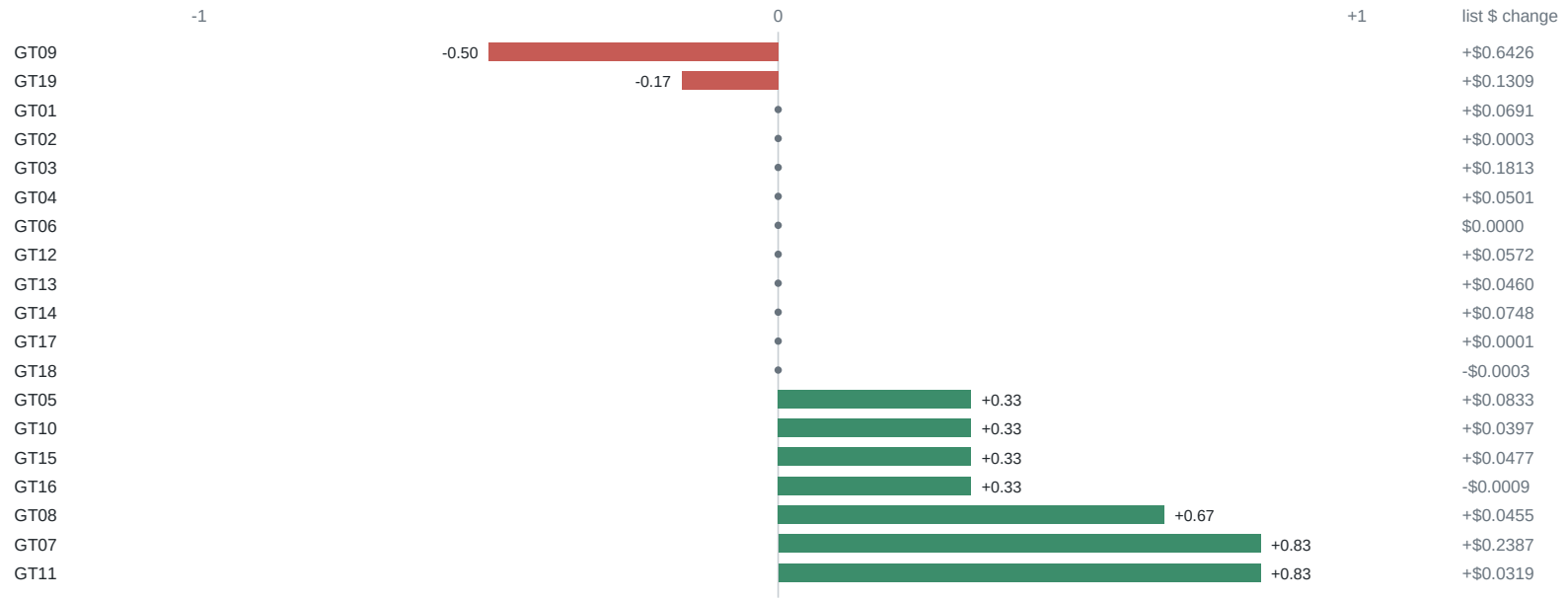
7.1 final, gpt-5.4 against 7.1 final



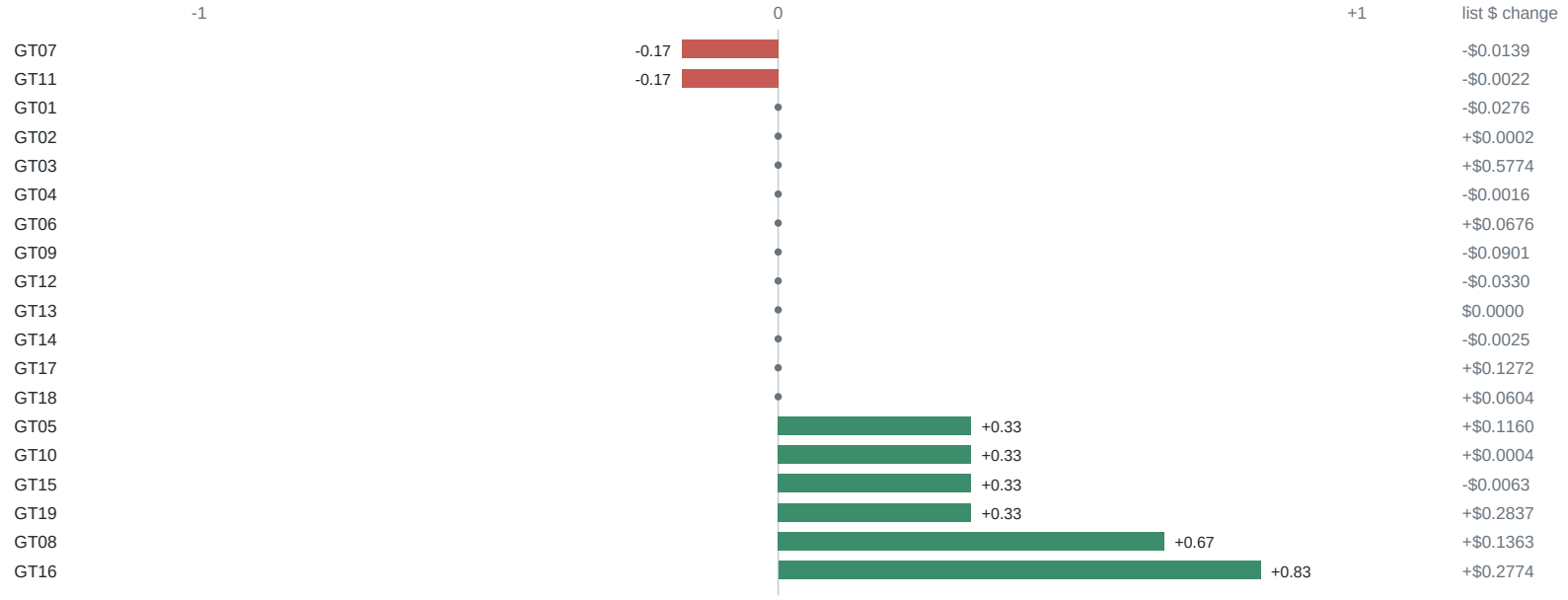
7.1 router by question type against 7.1 final



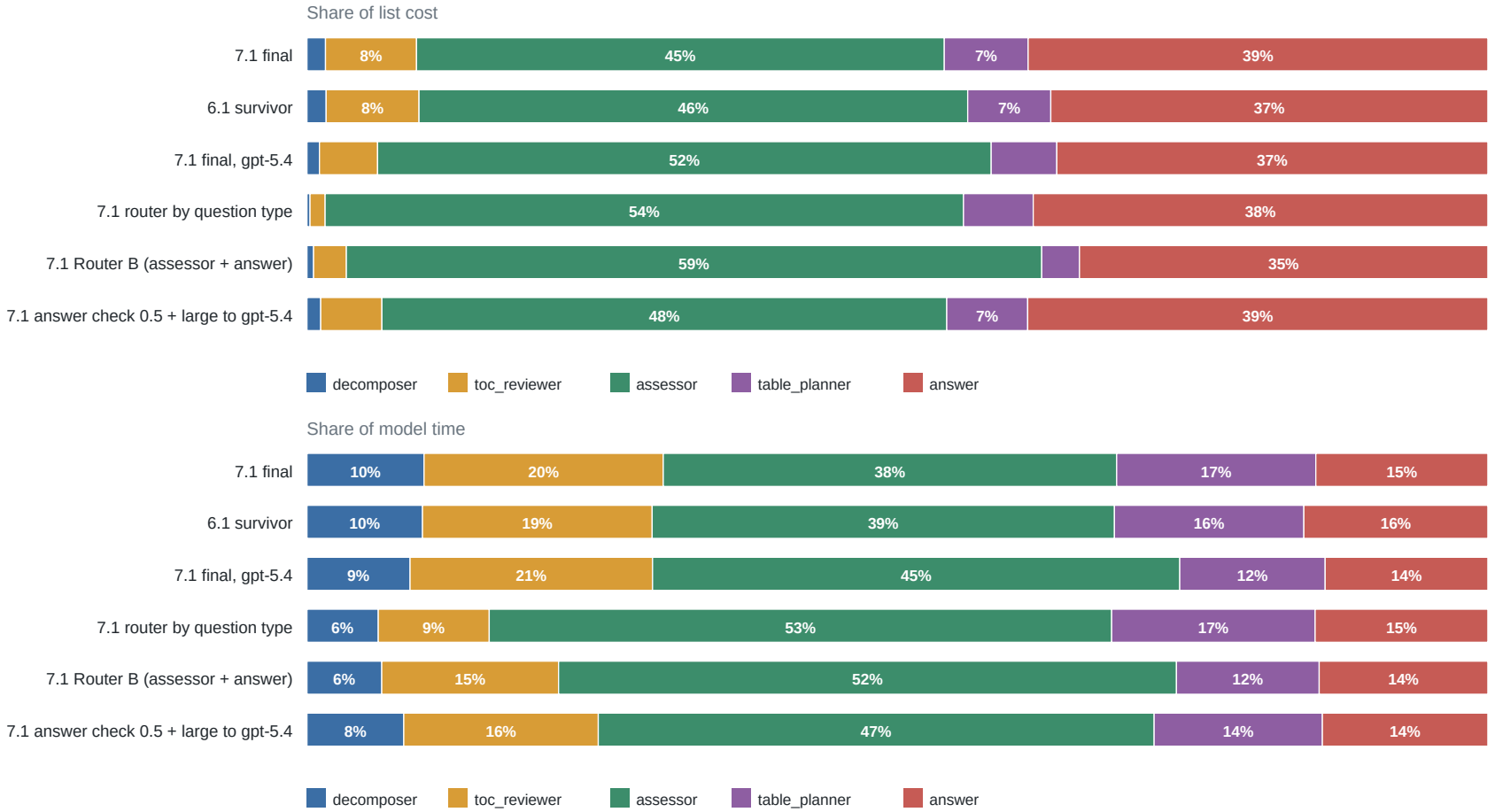
7.1 Router B (assessor + answer) against 7.1 final



7.1 answer check 0.5 + large to gpt-5.4 against 7.1 final



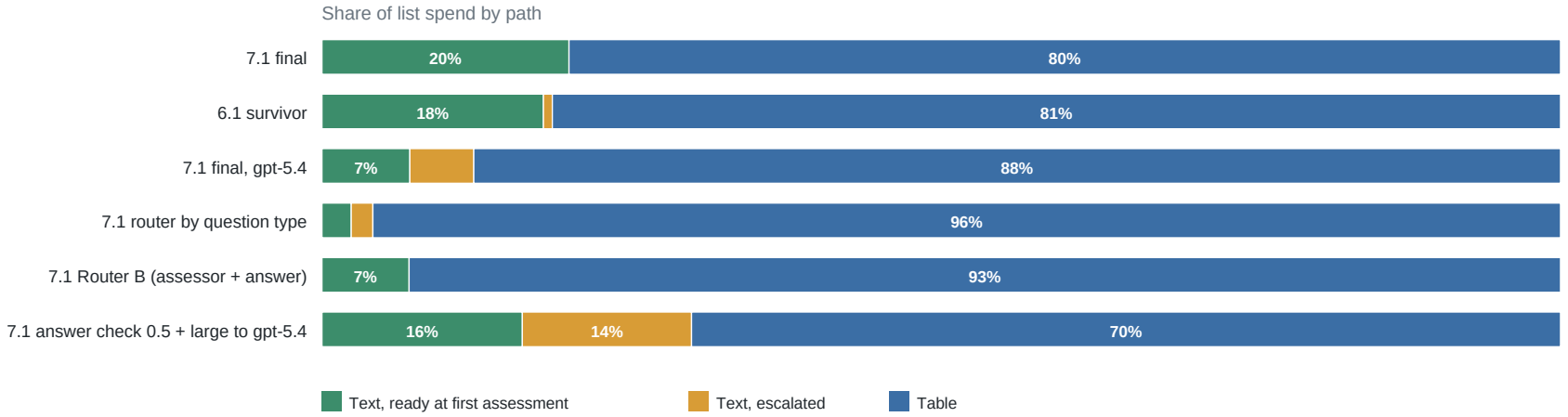
3. Where the cost goes: by role



Configuration	Role	Calls/q	Input tokens/q	Output tokens/q	Cached	List $/q	Share of list	Model s/q	Share of time	Median s/call
7.1 final	decomposer	1.00	0.4K	104	0%	$0.0008	2%	1.1	10%	1.04
7.1 final	toc_reviewer	1.00	3.0K	361	82%	$0.0039	8%	2.3	20%	2.20
7.1 final	assessor	2.02	26.9K	505	34%	$0.0224	45%	4.3	38%	1.89
7.1 final	table_planner	1.82	3.9K	140	47%	$0.0036	7%	1.9	17%	1.00
7.1 final	answer	1.00	25.1K	149	37%	$0.0195	39%	1.6	15%	1.52
6.1 survivor	decomposer	1.00	0.4K	105	0%	$0.0008	2%	1.0	10%	1.02
6.1 survivor	toc_reviewer	1.00	3.0K	353	55%	$0.0038	8%	2.1	19%	1.99

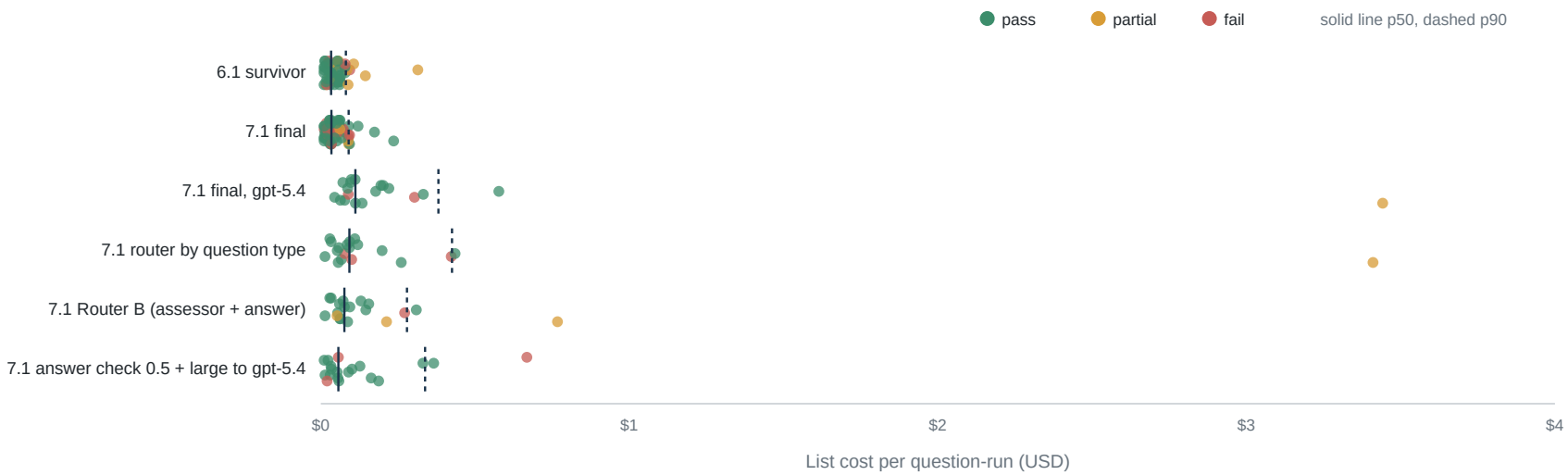
Configuration	Role	Calls/q	Input tokens/q	Output tokens/q	Cached	List $/q	Share of list	Model s/q	Share of time	Median s/call
6.1 survivor	assessor	2.09	26.9K	534	38%	$0.0226	46%	4.2	39%	1.85
6.1 survivor	table_planner	1.68	3.7K	137	38%	$0.0034	7%	1.7	16%	0.95
6.1 survivor	answer	1.00	22.8K	184	30%	$0.0180	37%	1.7	16%	1.42
7.1 final, gpt-5.4	decomposer	1.00	0.4K	114	0%	$0.0028	1%	1.9	9%	1.80
7.1 final, gpt-5.4	toc_reviewer	1.00	3.0K	351	0%	$0.0129	5%	4.5	21%	4.42
7.1 final, gpt-5.4	assessor	2.16	50.2K	683	35%	$0.1357	52%	9.7	45%	4.38
7.1 final, gpt-5.4	table_planner	1.95	5.0K	134	19%	$0.0145	6%	2.7	12%	1.34
7.1 final, gpt-5.4	answer	1.00	37.1K	168	0%	$0.0953	37%	3.0	14%	2.35
7.1 router by question type	decomposer	1.00	0.4K	103	0%	$0.0008	0%	1.2	6%	1.00
7.1 router by question type	toc_reviewer	1.00	3.0K	353	9%	$0.0038	1%	1.8	9%	1.76
7.1 router by question type	assessor	2.21	48.3K	735	33%	$0.1627	54%	10.3	53%	4.87
7.1 router by question type	table_planner	2.53	6.1K	175	15%	$0.0178	6%	3.4	17%	1.34
7.1 router by question type	answer	1.00	37.1K	205	0%	$0.1158	38%	2.8	15%	2.15
7.1 Router B (assessor + answer)	decomposer	1.00	0.4K	106	0%	$0.0008	1%	1.1	6%	1.03
7.1 Router B (assessor + answer)	toc_reviewer	1.00	3.1K	360	72%	$0.0039	3%	2.7	15%	2.24
7.1 Router B (assessor + answer)	assessor	2.21	30.1K	688	40%	$0.0834	59%	9.5	52%	4.06
7.1 Router B (assessor + answer)	table_planner	2.16	5.0K	175	14%	$0.0045	3%	2.2	12%	0.97
7.1 Router B (assessor + answer)	answer	1.00	26.0K	174	47%	$0.0490	35%	2.6	14%	2.15
7.1 answer check 0.5 + large to gpt-5.4	decomposer	1.21	0.5K	135	0%	$0.0015	1%	1.6	8%	1.13
7.1 answer check 0.5 + large to gpt-5.4	toc_reviewer	1.21	3.6K	429	70%	$0.0066	5%	3.2	16%	2.33
7.1 answer check 0.5 + large to gpt-5.4	assessor	2.79	34.6K	834	36%	$0.0608	48%	9.2	47%	2.53
7.1 answer check 0.5 + large to gpt-5.4	table_planner	2.53	5.7K	195	48%	$0.0087	7%	2.8	14%	1.07
7.1 answer check 0.5 + large to gpt-5.4	answer	1.21	26.8K	231	47%	$0.0495	39%	2.7	14%	1.63

4. By path: text ready, text escalated, table



Configuration	Path	Question-runs (share)	Quality	List $/q	Billed $/q	Seconds/q	Share of list spend
7.1 final	Text, ready at first assessment	15 (26%)	0.833	$0.0380	$0.0191	9.3	20%
7.1 final	Table	42 (74%)	0.667	$0.0545	$0.0404	15.4	80%
6.1 survivor	Text, ready at first assessment	13 (23%)	0.962	$0.0381	$0.0203	8.9	18%
6.1 survivor	Text, escalated	1 (2%)	1.000	$0.0200	$0.0200	8.2	1%
6.1 survivor	Table	43 (75%)	0.733	$0.0524	$0.0398	14.5	81%
7.1 final, gpt-5.4	Text, ready at first assessment	4 (21%)	1.000	$0.1142	$0.1142	13.5	7%
7.1 final, gpt-5.4	Text, escalated	1 (5%)	1.000	$0.3322	$0.3322	22.3	5%
7.1 final, gpt-5.4	Table	14 (74%)	0.821	$0.4033	$0.3219	27.3	88%
7.1 router by question type	Text, ready at first assessment	4 (21%)	1.000	$0.0338	$0.0332	8.9	2%
7.1 router by question type	Text, escalated	1 (5%)	0.000	$0.1000	$0.1000	11.3	2%
7.1 router by question type	Table	14 (74%)	0.821	$0.3916	$0.3160	26.5	96%
7.1 Router B (assessor + answer)	Text, ready at first assessment	5 (26%)	0.900	$0.0379	$0.0101	9.1	7%
7.1 Router B (assessor + answer)	Table	14 (74%)	0.857	$0.1787	$0.1339	24.9	93%
7.1 answer check 0.5 + large to gpt-5.4	Text, ready at first assessment	4 (21%)	1.000	$0.0980	$0.0235	11.1	16%
7.1 answer check 0.5 + large to gpt-5.4	Text, escalated	1 (5%)	1.000	$0.3310	$0.1949	25.0	14%
7.1 answer check 0.5 + large to gpt-5.4	Table	14 (74%)	0.786	$0.1214	$0.0901	26.0	70%

5. Tail and waste



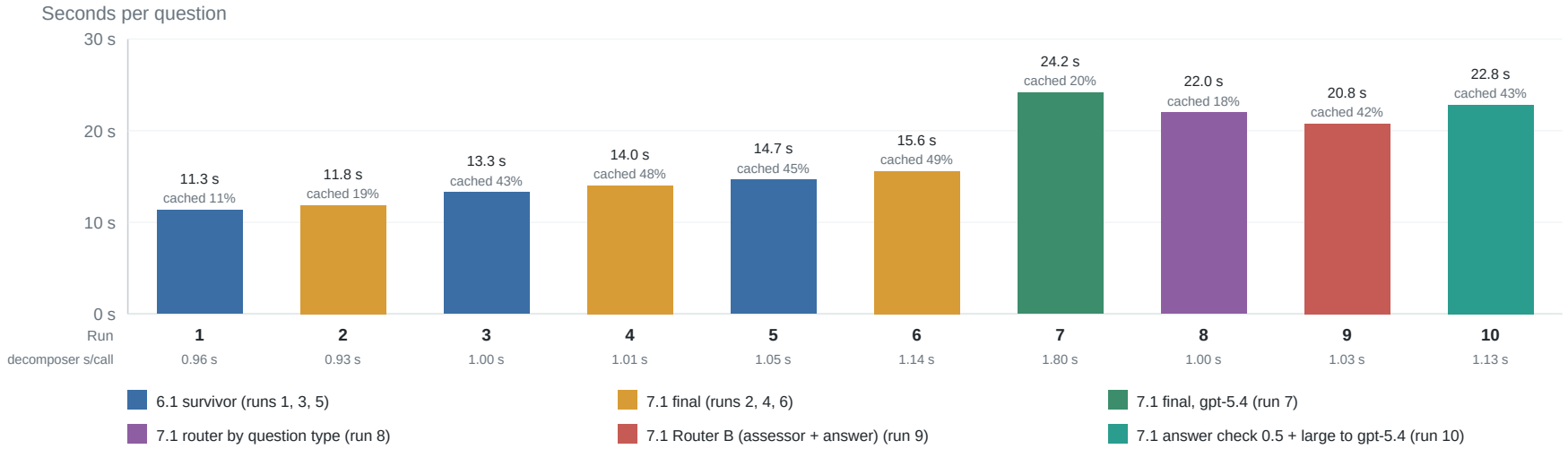
Configuration	List p50	List p90	List max	Max / mean	Seconds p50	Seconds p90	Seconds max	Spend on fails	Spend on partials	List $ per pass
7.1 final	$0.0343	$0.0902	$0.2360	4.7x	12.6	21.2	30.8	23%	14%	$0.0773
6.1 survivor	$0.0333	$0.0813	$0.3144	6.5x	11.8	20.1	30.5	13%	34%	$0.0692
7.1 final, gpt-5.4	$0.1120	$0.3812	$3.4416	10.2x	22.3	35.6	51.1	6%	53%	$0.4022
7.1 router by question type	$0.0927	$0.4252	$3.4103	11.3x	19.4	39.6	46.0	10%	60%	$0.3811
7.1 Router B (assessor + answer)	$0.0764	$0.2793	$0.7671	5.4x	21.2	32.6	46.4	10%	38%	$0.1794
7.1 answer check 0.5 + large to gpt-5.4	$0.0569	$0.3380	$0.6678	5.2x	15.7	57.0	79.1	31%	0%	$0.1514

Most expensive question-runs:

Configuration	Run	Question	Path	Verdict	List $	Seconds	Calls
7.1 final, gpt-5.4	adoring_hopper	GT09	Table	partial	$3.4416	42.0	12
7.1 router by question type	wizardly_yao	GT09	Table	partial	$3.4103	35.9	11
7.1 Router B (assessor + answer)	suspicious_liskov	GT09	Table	partial	$0.7671	30.8	10
7.1 answer check 0.5 + large to gpt-5.4	gifted_yao	GT03	Table	fail	$0.6678	62.1	20
7.1 final, gpt-5.4	adoring_hopper	GT07	Table	pass	$0.5770	51.1	15
7.1 router by question type	wizardly_yao	GT07	Table	pass	$0.4351	46.0	17
7.1 router by question type	wizardly_yao	GT03	Table	fail	$0.4228	38.0	11

Configuration	Run	Question	Path	Verdict	List $	Seconds	Calls
7.1 answer check 0.5 + large to gpt-5.4	gifted_yao	GT19	Table	pass	$0.3658	79.1	22
7.1 final, gpt-5.4	adoring_hopper	GT16	Text, escalated	pass	$0.3322	22.3	5
7.1 answer check 0.5 + large to gpt-5.4	gifted_yao	GT16	Text, escalated	pass	$0.3310	25.0	5

6. Comparability: run order, caching, and provider speed



Run (order)	Results CSV	Finished	Quality	List $/q	Billed $/q	Billed / list	Cached	Seconds per q	Decomp. s/call	TOC s/call	Assessor s/call	Planner s/call	Answer s/call
1. 6.1 survivor	eager_torvalds	2026-10-04 01:41	0.737	$0.0402	$0.0368	0.92	11%	11.3	0.96	1.80	1.58	0.93	1.30
2. 7.1 final	admiring_diffie	2026-10-04 01:45	0.658	$0.0507	$0.0432	0.85	19%	11.8	0.93	1.80	1.64	0.92	1.46
3. 6.1 survivor	hardcore_page	2026-10-04 01:50	0.816	$0.0570	$0.0372	0.65	43%	13.3	1.00	1.76	1.82	0.94	1.32
4. 7.1 final	eloquent_hoare	2026-10-04 01:55	0.711	$0.0459	$0.0286	0.62	48%	14.0	1.01	2.13	1.91	0.98	1.52
5. 6.1 survivor	confident_stallman	2026-10-04 02:00	0.816	$0.0485	$0.0310	0.64	45%	14.7	1.05	2.41	2.13	1.09	1.64
6. 7.1 final	confident_zuckerberg	2026-10-04 02:06	0.763	$0.0539	$0.0325	0.60	49%	15.6	1.14	2.52	2.08	1.10	1.63
7. 7.1 final, gpt-5.4	adoring_hopper	2026-10-04 10:12	0.868	$0.3387	$0.2787	0.82	20%	24.2	1.80	4.42	4.38	1.34	2.35
8. 7.1 router by question type	wizardly_yao	2026-10-04 13:03	0.816	$0.3009	$0.2451	0.81	18%	22.0	1.00	1.76	4.87	1.34	2.15
9. 7.1 Router B (assessor + answer)	suspicious_liskov	2026-10-04 13:11	0.868	$0.1417	$0.1013	0.72	42%	20.8	1.03	2.24	4.06	0.97	2.15

Run (order)	Results CSV	Finished	Quality	List $/q	Billed $/q	Billed / list	Cached	Seconds per q	Decomp. s/call	TOC s/call	Assessor s/call	Planner s/call	Answer s/call
10. 7.1 answer check 0.5 + large to gpt-5.4	gifted_yao	2026-10-04 13:31	0.842	$0.1275	$0.0816	0.64	43%	22.8	1.13	2.33	2.53	1.07	1.63

Run times from: run_manifest.csv.

Warnings

- Single run: 7.1 final, gpt-5.4, 7.1 router by question type, 7.1 Router B (assessor + answer), 7.1 answer check 0.5 + large to gpt-5.4. One run cannot show a difference smaller than the run-to-run spread of a repeated configuration.

- 6.1 survivor: list cost per question ranges from $0.0402 to $0.0570 across its own runs (42% apart).

- Cached share ranges from 11% to 49% across runs (it grows with repeats of the same questions). Billed cost depends on run order; compare list cost.

- The decomposer's median call took 0.93-1.80 s across runs (93% apart). Its prompt is the same in every configuration, so time differences of this size are the provider's speed, not the system.

7. Safeguard markers

Configuration	Question-runs	Evidence notice	Noticed: pass / partial / fail	Grounding checked	Flagged	Flagged: pass / partial / fail	Check passed, judge failed	Input sanitized
7.1 final	57	17 (30%)	11/4/2	57	10 (18%)	8/1/1	12	0
6.1 survivor	57	17 (30%)	12/4/1	not checked	-	-	-	0
7.1 final, gpt-5.4	19	5 (26%)	4/1/0	19	1 (5%)	1/0/0	2	0
7.1 router by question type	19	6 (32%)	5/1/0	19	5 (26%)	3/1/1	2	0
7.1 Router B (assessor + answer)	19	9 (47%)	7/2/0	19	2 (11%)	2/0/0	1	0
7.1 answer check 0.5 + large to gpt-5.4	19	7 (37%)	6/0/1	19	3 (16%)	2/0/1	2	0

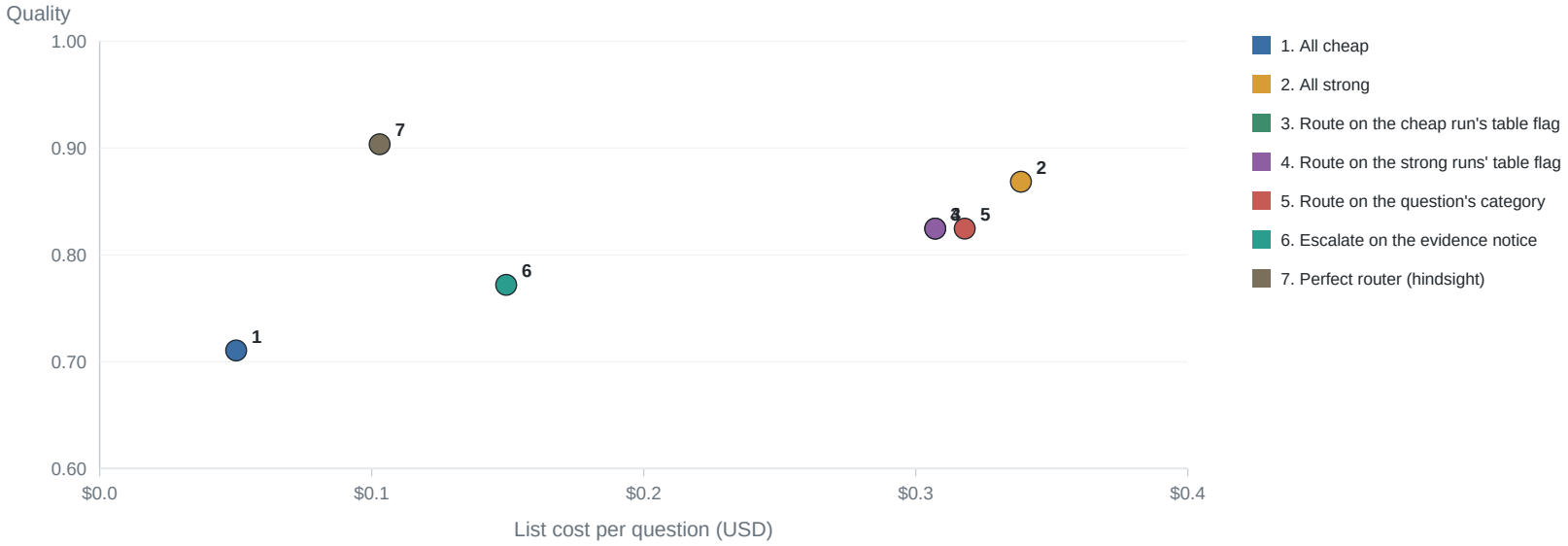
A flagged answer the judge passed is a false alarm the user still sees (policy 'surface' labels it). 'Check passed, judge failed' counts wrong answers the deterministic check cannot see.

Spans the grounding check could not find in the evidence:

Span	Kind	Times	Judge: pass / partial / fail	Configurations
Russian Empire	name	6	6/0/0	7.1 Router B (assessor + answer); 7.1 answer check 0.5 + large to gpt-5.4; 7.1 final; 7.1 final, gpt-5.4; 7.1 router by question type
Russia-related Olympic	name	3	3/0/0	7.1 final
Spanish armies, led by Fernando Álvarez de Toledo, 3rd Duke of Alba, defending the claim of King Philip II of Spain to the Portuguese throne, defeat the armies of Portuguese claimant António, Prior of Crato	quotation	3	3/0/0	7.1 Router B (assessor + answer); 7.1 final; 7.1 router by question type
Winter Olympic Games Sport	name	2	2/0/0	7.1 final
Barack Obama.html	name	1	0/0/1	7.1 router by question type
Bill Clinton.html	name	1	0/0/1	7.1 router by question type
Summer Olympic Games Sport	name	1	0/0/1	7.1 final
To Barack Obama	name	1	0/1/0	7.1 final
To Bill Clinton	name	1	0/1/0	7.1 final
a final combined computation across all relevant tables that normalizes actor names and identifies which actor or actors have the highest number of wins from 2000 through 2020 inclusive is not supplied	quotation	1	1/0/0	7.1 router by question type
could not be found or computed	quotation	1	0/1/0	7.1 router by question type
did not produce a usable result	quotation	1	1/0/0	7.1 answer check 0.5 + large to gpt-5.4
every person who was nominated at all three ceremonies	quotation	1	0/0/1	7.1 answer check 0.5 + large to gpt-5.4

8. Routing

Cheap: 7.1 final. Strong: 7.1 final, gpt-5.4. Text categories for the category policy: Direct fact; Direct fact and quotation; Process explanation; Inference limitation; Diagnostic reasoning and limitations; People connection: intermediary person; People connection: shared work and roles.



Policy	Rule	Quality	List $/q	Seconds/q	Sent to strong	Paid twice	Saving vs all strong
1. All cheap	every question on the cheap configuration	0.711	$0.0502	13.8	0%	0%	85%
2. All strong	every question on the strong configuration	0.868	$0.3387	24.2	100%	0%	0%
3. Route on the cheap run's table flag	strong when the cheap run's TOC review flagged a table	0.825	$0.3072	22.6	74%	0%	9%
4. Route on the strong runs' table flag	strong when most strong runs flagged a table for the question	0.825	$0.3072	22.6	74%	0%	9%
5. Route on the question's category	strong unless the category is one of the text categories	0.825	$0.3180	23.0	79%	0%	6%
6. Escalate on the evidence notice	cheap first; rerun on strong when the evidence notice fired (both paid)	0.772	$0.1494	22.5	30%	30%	56%
7. Perfect router (hindsight)	strong only where the strong mean scores higher; the lower bound for any router	0.904	$0.1029	17.9	25%	0%	70%

Simulated on saved runs: a routed question takes the strong configuration's mean outcome, cost, and time for that question. A routed system is a different run, so the simulation predicts and a live run confirms.

Configuration	Route	Question-runs (share)	Quality	List $/q	Seconds/q
7.1 router by question type	cheap	5 (26%)	0.800	$0.0470	9.4
7.1 router by question type	strong	14 (74%)	0.821	$0.3916	26.5

Configuration	Route	Question-runs (share)	Quality	List $/q	Seconds/q
7.1 Router B (assessor + answer)	cheap	5 (26%)	0.900	$0.0379	9.1
7.1 Router B (assessor + answer)	strong	14 (74%)	0.857	$0.1787	24.9
7.1 answer check 0.5 + large to gpt-5.4	cheap	11 (58%)	0.818	$0.0354	13.6
7.1 answer check 0.5 + large to gpt-5.4	escalated	4 (21%)	0.750	$0.3309	55.9
7.1 answer check 0.5 + large to gpt-5.4	strong	4 (21%)	1.000	$0.1773	15.1